Wentao Zhang

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- Curriculum vitae prepared for nomination to the *Electronics* Young Investigator Award 2026.

Research Focus

I develop **physics-informed artificial intelligence and real-time digital twins for power-electronic and electromechanical systems**. My work couples deep-learning surrogate models, multiphysics simulation, and hardware measurement to make the design and operation of electric drives faster, more accurate, and more energy-efficient. The methods are device-agnostic and apply across power electronics, electric machines, and industrial electronic systems.

Areas of Expertise

- **Artificial intelligence and machine learning.** Physics-informed and operator-learning neural networks; deep-learning surrogate modelling of electromagnetic and power-electronic fields; differentiable models for gradient-based design; PyTorch with GPU computing.
- **Power electronics and electric drives.** Three-phase converter prototyping, PWM excitation and loss modelling, real-time embedded control (TI C2000, STM32), torque-speed and efficiency-map characterisation.
- **Digital twins and reduced-order modelling.** High-fidelity lumped-parameter thermal networks, fluid-solid coupled cooling models, hardware-validated real-time twins of rotating electrical machines.
- **Multiphysics modelling and design.** Coupled electromagnetic, thermal, fluid, and structural-mechanical analysis; in-house finite-element code; multi-objective design optimisation; design for manufacture.
- **Engineering software.** JMAG, ANSYS Maxwell, Motor-CAD, ANSYS Mechanical, MATLAB / Simulink. **Programming.** Python (primary), C / C++, MATLAB, Rust.
- **Experimental and hardware.** AVL dynamometer testing, multi-thermocouple instrumentation, converter and prototype build, manufacturing-drawing and tolerance release.

Degrees

Southeast University 2021/09 - 2025/09
Ph.D. in Electrical Engineering Nanjing, China
- Dissertation, “Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles”. Doctoral supervisor Prof. Zhongze Wu.

Southeast University 2018/09 - 2021/06
M.Eng. in Electrical Engineering Nanjing, China
- Dissertation, “Analysis and Design of High-Performance Wound-Excited Magnetic Field Modulation-based Brushless Machines”.
- 1st Prize, IEEE IAS CMD Thesis Contest, 2023 (awarded for this master’s thesis).
- Outstanding Master Degree Thesis Award, Jiangsu Province, 2022.

Southeast University 2014/09 - 2018/06
B.Eng. in Electrical Engineering Nanjing, China
- Dissertation, “Comparative Study of Control Systems for PM Machine in Electric Vehicles”.

Current Employment

University of Nottingham 2025/09 - present
Research Assistant, Power Electronics and Machines Centre (PEMC) Nottingham, UK
- Within the EPSRC-funded AIMD programme (Nottingham, Bristol, Newcastle), I work on next-generation magnet-free synchronous machines, develop an in-house finite-element code package, and extend operator-learning neural surrogates to accelerate electric-drive design.

Previous Work Experience

Teaching Assistant, Southeast University 2021 - 2025
Postgraduate course "Integrated Practice of Electrical Machine Systems" Nanjing, China

- Prepared and delivered tutorial slides, ran daily Q&A and consultation hours, reviewed student final-project proposals with structured written feedback, and provided mark recommendations to the course coordinator.

Research Funding and Grants

- I hold no funding as principal investigator to date, consistent with my early-career stage. I have contributed as a **participating researcher** to externally funded programmes, including the EPSRC **AIMD** programme at the University of Nottingham (2025 to present) and the **Inovance**-funded industrial traction-machine development programme at Southeast University (2021 to 2025).

Research Output

My record comprises **23 journal papers** (19 in IEEE Transactions; 11 first-author), **18 peer-reviewed conference papers** (8 first-author), and **9 patents**, on physics-informed AI, digital twins, and multiphysics design of power-electronic and electromechanical systems. The full list follows.

Journal Articles (23 papers, 19 in IEEE Transactions)

[J1] **Wentao Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC Winding Induced Voltage in Wound-Field Flux-Switching Machine With Air-Gap Field Modulation Principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J2] **Wentao Zhang**, Z. Wu, W. Hua, Z. Zhu, Z. Wu, J. Hu, et al., "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth Tips," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1976–1990, Mar.–Apr. 2022.

[J3] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 3966–3977, Jul.–Aug. 2023.

[J4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **Wentao Zhang**, Z. Wu and W. Hua, "Design of *n*-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025. (2025 IEEE TTE Prize Paper Award, 2nd Prize.)

[J6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025.

[J7] **Wentao Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, et al., "Design Tradeoff Between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

[J8] **Wentao Zhang**, Y. Fan, Z. Zhu, Z. Wu, W. Hua, and M. Cheng, "Analysis of DC Winding Induced Voltage in Wound-Rotor Synchronous Machines by Using the Air-Gap Field Modulation Principle," *World Electr. Veh. J.*, vol. 13, no. 11, Art. no. 215, 2022.

[J9] **Wentao Zhang**, Z. Wu, Z. Cang, W. Hua, Y. Fan, and M. Cheng, "Influence of End-Effect in Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 785–798, Jan. 2026.

[J10] **Wentao Zhang**, Z. Wu and W. Hua, "High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] **Wentao Zhang**, Z. Wu and W. Hua, "Design of Notching and Mechanical Stress Relieving Grooves for High-Speed PM Machine in EVs," *IEEE Trans. Ind. Appl.*, early access, 2026.

[J12] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "A Transient-State Lumped Parameter Thermal Model for Brushless Wound Field Switched Flux Machines," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J13] Z. Wu, Z. Zhu, C. Wang, W. Hua, K. Wang, and **Wentao Zhang**, "Influence of Rotor Iron Bridge Position on DC-Winding-Induced Voltage in Wound Field Switched Flux Machine Having Partitioned Stators," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 20–28, Sep. 2021.

[J14] Z. Wu, Z. Zhu, S. Cai, W. Hua, and **Wentao Zhang**, "Enhancement of Torque Density in Wound Field Switched Flux Machines With Partitioned Stators Using Assisted Ferrites," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 42–51, Sep. 2021.

[J15] Z. Wu, L. Jin, W. Hua, **Wentao Zhang**, and M. Cheng, "Harmonics Orders Modeling of DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines Under PWM Excitation," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 3, pp. 7040–7050, Sep. 2024.

[J16] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 60, no. 1, pp. 460–476, Jan.–Feb. 2024.

[J17] Z. Wu, Z. Zhu, W. Hua, S. Akehurst, X. Zhu, **Wentao Zhang**, et al., "Analysis and Suppression of Induced Voltage Pulsation in DC Winding of Five-Phase Wound-Field Switched Flux Machines," *IEEE Trans. Energy Convers.*, vol. 34, no. 4, pp. 1890–1905, Dec. 2019.

[J18] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang**, and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2874–2883, Mar.–Apr. 2025.

[J19] X. Yan, Z. Wu, W. Hua, **Wentao Zhang**, and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6811–6822, Sep.–Oct. 2024.

[J20] H. Hua, X. Chen, D. Li, **Wentao Zhang**, Z. Wu, and W. Hua, "Investigation of Torque Improvement of Modular Stator Surface-Mounted Permanent Magnet Machines With Flux Gaps," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6787–6798, Sep.–Oct. 2024.

[J21] J. Chen, W. Hua, L. Shao, Z. Wu, X. Yan, and **Wentao Zhang**, "Modified Magnetic Equivalent Circuit of Double-Stator Single-Rotor Axial Flux Permanent Magnet Machine Considering Stator Radial-End Flux-Leakage," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 195–207, Feb. 2024.

[J22] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang**, and W. Hua, "Influence of PWM Excitation on Loss and Temperature of Flat Wire Permanent Magnet Machines for Electric Vehicles," *IEEE Trans. Ind. Electron.*, vol. 73, no. 4, pp. 5174–5185, Apr. 2026.

[J23] Z. Wu, E. Fu, J. Du, **Wentao Zhang**, and W. Hua, "Quantitative Comparison of Electromagnetic Vibration of PMSMs With Glued and Welded Stator Core for Electric Vehicles," *IEEE Trans. Ind. Appl.*, early access, 2025.

Conference Papers (18 papers)

[C1] **Wentao Zhang**, Z. Wu and W. Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," in *Proc. Int. Conf. Elect. Mach. (ICEM)*, Funchal, Madeira, Portugal, 2026. (Accepted.)

[C2] **Wentao Zhang**, Z. Wu, W. Hua and Z. Zhu, "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth-Tips," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, Gothenburg, Sweden, 2020, pp. 522–528.

[C3] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," in *Proc. IEEE Energy Convers. Congr. Expo.*, Detroit, MI, USA, 2022, pp. 1–7.

[C4] **Wentao Zhang**, Z. Wu, W. Hua, Y. Fan, Z. Xu and M. Cheng, "Design Tradeoff between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," in *Proc. IEEE Energy Convers. Congr. Expo.*, Nashville, TN, USA, 2023, pp. 5060–5066.

[C5] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," in *Proc. Int. Elect. Energy Conf.*, Hefei, China, 2023, pp. 3371–3376. (Best Paper Award.)

[C6] **Wentao Zhang**, Z. Wu, W. Hua, Y. Fan and M. Cheng, "Influence of the End-Effect in Wound Field Switched Flux Machine," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 1223–1227.

[C7] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," in *Proc. Int. Conf. Energy Technol. Future Grids*, Wollongong, Australia, 2023, pp. 1–6.

[C8] **Wentao Zhang**, Z. Wu, W. Hua and J. Qi, "Analysis of PM Volume Reduction Methods for Hybrid Excitation Switched Flux Machines," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2024, pp. 3152–3157.

[C9] **Wentao Zhang**, Z. Wu and W. Hua, "Field-Oriented Design of Notching and Mechanical Stress Relieving Grooves for Rotors in Flat Wire Permanent Magnet Machines," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 304–309.

[C10] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage in Wound Field Switched Flux Machines," in *Proc. IEEE Energy Convers. Congr. Expo.*, Detroit, MI, USA, 2022, pp. 1–7.

[C11] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang** and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 442–447.

[C12] M. P. Koroma, W. Hua, Z. Wu, L. Jin and **Wentao Zhang**, "Influence of Magnet Layer Numbers on Electromagnetic Performance of Interior Permanent Magnet Machines," in *Proc. Int. Conf. Elect. Mach. Syst.*, Zhuhai, China, 2023, pp. 696–700.

[C13] X. Chen, H. Hua, D. Li, **Wentao Zhang**, Z. Wu and W. Hua, "Modular Stator Surface-Mounted Permanent Magnet Machines with Flux Gaps," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, San Francisco, CA, USA, 2023, pp. 1–7.

[C14] X. Yan, Z. Wu, W. Hua, **Wentao Zhang** and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, San Francisco, CA, USA, 2023, pp. 1–6.

[C15] Z. Wu, L. Jin, **Wentao Zhang**, X. Yan, H. Ge, H. Hua, W. Hua and M. Cheng, "A Lumped Parameter Thermal Model for Variable Flux Reluctance Machine Considering Relative Position of DC and AC Windings," in *Proc. Int. Conf. Elect. Mach.*, Torino, Italy, 2024, pp. 1–7.

[C16] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang** and W. Hua, "Quantitative Analysis of the Torque-Speed Curve for a Flat Wire Permanent Magnet Machine Drive System," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 40–44.

[C17] Z. Wu, E. Fu, J. Du, **Wentao Zhang** and W. Hua, "Influence of Stator Core on the Vibration Behavior of Permanent Magnet Machines with Hairpin Winding," in *Proc. IEEE Int. Elect. Mach. Drives Conf.*, 2025, pp. 34–39.

[C18] Z. Wu, **Wentao Zhang**, Z. Wang, L. Shao, W. Hua and M. Cheng, "Stator Semi-Skewing Method for High-Speed Permanent Magnet Machine," in *Proc. Int. Conf. Elect. Mach. Syst.*, 2025, pp. 693–697.

Patents (9)

[P1] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Inductive Ripple Voltage Suppression Method Based on Low-Pass Filter Circuit," CN118508832A, 2024.

[P2] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Disk Brushless Electric Excitation Magnetic Field Modulation Motor With Double-Row Stator Structure," CN118100572A, 2024.

[P3] Z. Wu, **Wentao Zhang**, H. Huang, and W. Hua, "Method for Constructing a High-Precision Motor Performance Database Based on a Dual-Model Strategy," CN117972924A, 2024.

[P4] Z. Wu, **Wentao Zhang**, W. Hua, and Z. Zou, "Method for Establishing General Lumped Parameter Thermal Network Model of Motor," CN117094088A, 2023.

[P5] Z. Wu, **Wentao Zhang**, L. Jin, and W. Hua, "Method for Analyzing Harmonic Orders of Induced Ripple Voltage of Excitation Winding," CN115642847A, 2023.

[P6] Z. Wu, **Wentao Zhang**, J. Liao, Y. Fan, W. Hua, and M. Cheng, "Excitation Winding Induced Pulse Voltage Suppression Method Based on Harmonic Current Injection," CN115208245A, 2022.

[P7] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "High-Precision Field Path Coupling Design Method for Brushless Electric Excitation Magnetic Field Modulation Motor," CN114781227A, 2022.

[P8] Z. Wu, **Wentao Zhang**, W. Hua, and Z. Zhu, "Method for Weakening the Induced Pulsating Voltage of a Brushless Electric Excitation Magnetic Field Modulated Motor," CN114189185A, 2022.

[P9] W. Hua, **Wentao Zhang**, Z. Wu, and S. Ding, "Method for Reducing Torque Ripple and Induced Voltage of Electric Excitation Flux Switching Motor," CN111769697A, 2020.

Awards and Honours

- 2025 **1st Prize, IEEE TEC Electric Vehicle Design Competition**.
- 2025 **2nd Prize Paper Award, IEEE Transactions on Transportation Electrification**.
- 2024 2nd Prize, 11th Energy Equipment Innovative Design Competition for China Postgraduate.
- 2023 CIEEC Best Paper Award.
- 2023 1st Prize, IEEE IAS CMD Thesis Contest.
- 2023 Wiscom System Scholarship.
- 2023 1st Prize Academic Scholarship.
- 2022 Outstanding Master Degree Thesis Award, Jiangsu Province.

Academic Service and Leadership

- **Referee** for IEEE Transactions journals, including *IEEE Trans. Industry Applications*, *IEEE Trans. Energy Conversion*, and *IEEE Trans. Transportation Electrification.*

- **Co-Organiser and Co-Chair**, ICEM 2026 Special Session "Next-Generation Electrified Transport: Advances in Electric Drive Systems," Funchal, Portugal, Sep. 2026 (with Dr Lingyun Shao, NUAA).
- Provided day-to-day technical guidance to junior PhD and M.Eng. researchers in multiphysics modelling, finite-element analysis, and PWM-loss coupling, and acted as **technical lead** on the two-prototype Inovance development programme.

Scientific Impact and Innovation

- **Internationally recognised research.** My first-author paper on *n*-layer flat-wire winding design [J5] received the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**. In the same year my electric-drive design entry won **1st Prize in the IEEE TEC Electric Vehicle Design Competition**, an international contest organised by the IEEE Transactions on Energy Conversion.
- **Innovation and technology translation.** 9 patents, and two industrially deployed high-speed oil-cooled traction-machine prototypes delivered with **Inovance** through full design-build-test cycles (measured up to 270 kW, 7.87 kW/kg, and 97.4% peak efficiency at 20,000 rpm), contributing to energy-efficient transport electrification.
- **Open science (planned).** Intention to release the in-house finite-element code, structured simulation datasets, and trained surrogate models, to support reproducible research.

Language Skills

- **Mandarin Chinese**, native language. **English**, full professional proficiency (the medium of all 41 peer-reviewed publications and of my current research and employment in the United Kingdom).

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